

Quantum Category $\mathcal{O}$, III

Images of Vermas

Recap and Questions

Last time we have stated the main equivalence between triangulated categories

$$D^b(\mathcal{O}_\varepsilon^\Theta) \xrightarrow{\sim} {}_gH := K^b({}_g\underline{SMod})$$

Questions

1. Where does $\mathcal{O}_\varepsilon^\Theta \subset D^b(\mathcal{O}_\varepsilon^\Theta)$ go under $(*)$?
2. Where do Verma modules $\Delta_\varepsilon(\lambda) \in \mathcal{O}_\varepsilon^\Theta, \lambda \in \Theta$, go?

From 2) to 1)

It turns out that we can, to some extent, answer 1) if we know an answer to 2). Define the following subcategories in $D^b(\mathcal{O}_\varepsilon^\Theta)$:

$$\mathcal{T}^{>0} = \{X \in D^b(\mathcal{O}_\varepsilon^\Theta) \mid \text{Hom}_{D^b(\mathcal{O}_\varepsilon^\Theta)}(\Delta_\varepsilon(\lambda)[i], X) = 0, \forall \lambda \in \Theta, i \geq 0\}$$

$$\mathcal{T}^{\leq 0} = \{Y \in D^b(\mathcal{O}_\varepsilon^\Theta) \mid \text{Hom}_{D^b(\mathcal{O}_\varepsilon^\Theta)}(Y, X) = 0 \forall X \in \mathcal{T}^{>0}\}$$

Exercise 1. Show that $\mathcal{O}_\varepsilon^\Theta = \mathcal{T}^{\leq 0} \cap \mathcal{T}^{>0}[1]$ (hint: $\forall M \in \mathcal{O}_\varepsilon^\Theta \exists \lambda \in \Theta$ s.t. $\text{Hom}_{\mathcal{O}_\varepsilon^\Theta}(\Delta_\varepsilon(\lambda), M) \neq 0$).

Affine Braid Group Action

Motivation: $\mathcal{O}_\varepsilon^\Theta$ has very easy categorical symmetry – the shift of grading by an element of $d\Lambda_0$ sending Vermas to Vermas. Can we see that for ${}_gH$?

To W^a we can assign the corresponding affine braid group Br^a : it's generated by $T_s, s \in S$, with relations

$$\underbrace{T_s T_t T_s \dots}_{m_{st}} = \underbrace{T_t T_s T_t \dots}_{m_{st}} \text{ for } s \neq t, \quad m_{st} = \text{order of } st.$$

We want to construct an action of Br^a on a deformed version of H by equivalences (of triangulated categories), which is more elementary than dealing w. ${}_gH$ or H .

- Deformed version: $\widehat{R} = \varprojlim_n S(v) / (v)^n$, completion at 0

$\underline{\text{SBim}}^\wedge$: full subcategory in $\widehat{R}$ -bimod w. objects $B \otimes_R \widehat{R} \simeq \widehat{R} \otimes_R B$

w. $B \in \underline{\text{SBim}}$, monoidal additive category

$\rightarrow \widehat{H} = K^b(\underline{\text{SBim}}^\wedge)$ w. monoidal structure denoted by $*$

$$\text{ev}_0 : \underline{\text{SBim}}^\wedge \rightarrow \underline{\text{SMod}}, \quad B \mapsto B / BV \sim \text{ev}_0 : \widehat{H} \rightarrow H$$

- Br^a -action (Rouquier): $\exists$ group homomorphism

$\mathcal{R} : \text{Br}^a \rightarrow \text{group of iso. classes of invertible objects in } \widehat{H}$

$$T_s \mapsto \left[0 \rightarrow \widehat{R} \xrightarrow[\deg -1]{1 \mapsto \alpha_s \otimes 1 + 1 \otimes \alpha_s} \widehat{R} \otimes_{\widehat{R}^s} \widehat{R} \rightarrow 0 \right]$$

Objects $\widehat{\Delta}(x)$ and $\Delta(x)$

$$x \in W^a \sim T_x \in \text{Br}^a \quad (x = s_1 \dots s_e, \text{ reduced, } \sim T_x := T_{s_1} \dots T_{s_e})$$

$$\widehat{\Delta}(x) := \mathcal{R}(T_{x^{-1}}^{-1}) \in \widehat{H}, \quad \Delta(x) := \text{ev}_0(\widehat{\Delta}(x)) \in H.$$

Remark 1. One can play the same game as in Exercise in Sec 1.1 w. the objects $\Delta(x) \in \mathcal{H}$ getting an abelian subcategory in $\mathcal{H}$ (the heart of a t -structure, in fact). In the constructible realization of $\mathcal{H}$, this is the category of perverse sheaves.

Lattice and Stabilization

The embedding $\Lambda_0 \hookrightarrow W^a$ lifts to $\Lambda_0 \rightarrow \text{Br}^a$ uniquely so that for $\lambda \in \Lambda_0^+$ (dominant) $\lambda \mapsto T_{t_\lambda}$ (unique b/c $\forall$ element of Λ_0 is the difference of dominant ones).

Notation: $J_\mu := \text{image of } \mu \in \Lambda_0 \text{ in } \text{Br}$.

Note that $\widehat{\Delta}(xt_\mu) \not\simeq \widehat{\Delta}(x)^* J_\mu$ for general x, μ . But there's "stabilization."

Observation 1. $\lambda, \mu \in -\Lambda_0^+, w \in W, x = wt_\lambda \Rightarrow \ell(xt_\mu) = \ell(x) + \ell(t_\mu)$

$$\Rightarrow T_{(xt_{-\mu})^{-1}}^{-1} J_{-\mu} = T_{x^{-1}}^{-1} \Rightarrow \widehat{\Delta}(xt_{-\mu}) = \widehat{\Delta}(x)^* J_{-\mu}$$

Lemma 1 (Definition of $\widehat{\Delta}^{st}(x)$). $\forall x \in W^a \exists \mu_x \in \Lambda_0^+$ s.t. $\widehat{\Delta}(xt_{-\mu})^* J_\mu$ doesn't depend on μ if $\mu \geq \mu_x$.

Denote this object by $\widehat{\Delta}^{st}(x)$ ("st" for stabilized).

Exercise 2. $\widehat{\Delta}^{st}(xt_\mu) \simeq \widehat{\Delta}^{st}(x)^* J_\mu \forall x \in W^a, \mu \in \Lambda_0$.

$$\Delta^{st}(x) := \text{ev}_0 \left(\widehat{\Delta}^{st}(x) \right) \in \mathcal{H}$$

Theorem 1. Let $\Theta = W^a \cdot 0$. $\exists$ equivalence $D^b(O_\varepsilon^\Theta) \xrightarrow{\sim} \mathcal{H}$ w. $\Delta_\varepsilon(x^{-1} \cdot (-2\rho)) \mapsto \Delta^{st}(x)$.

Singular Blocks

$\mathbb{A}$: anti-dominant alcove in Λ_0 , $\lambda^\circ \in \Theta \cap \mathbb{A} \rightsquigarrow J \subset S \rightsquigarrow R^J \subset R \sim \pi_J: \underline{SMod} \rightarrow_J \underline{SMod} \sim \pi_J: H \rightarrow_J \mathcal{H}$.

Observation 2. $\pi_J(\Delta(wx)) \simeq \pi_J(\Delta(x)) \quad \forall w \in W_J, x \in W^a$.

Compare to: Behavior of Vermas in the BGG category $\mathcal{O}$ under translations to the wall.

From Observation (& its deformed version): $\pi_J(\Delta^{st}(wx)) \simeq \pi_J(\Delta^{st}(x))$

Theorem 2. Let $\Theta = W^a \cdot 0$. $\exists$ equivalence $D^b(O_\varepsilon^\Theta) \xrightarrow{\sim} {}_J\mathcal{H}$ w. $\Delta_\varepsilon(x^{-1} \cdot \lambda^\circ) \mapsto \pi_J(\Delta^{st}(x))$.